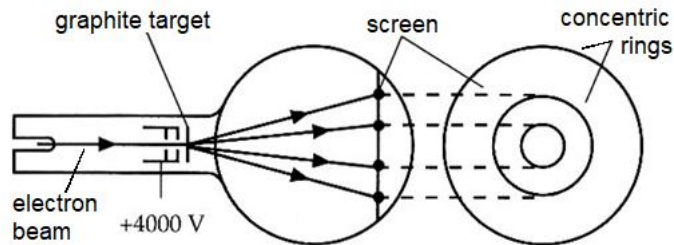


- 28** In an experimental setup, a beam of accelerated electrons strikes an extremely thin layer of polycrystalline graphite target and concentric rings are seen on the tube face as shown.



If the setup is modified such that instead of electron beam, a beam of protons strike the graphite target with the same speed, the radii of the concentric rings will

- A** decrease.
- B** increase.
- C** stay the same.
- D** not be seen. (No rings appear.)

A proton with the same v but bigger m than electron will have a much larger momentum p

de Broglie relation: $\lambda = \frac{h}{p}$ hence wavelength of proton is smaller

hence radii decrease.

29 The graph shows how the binding energy per nucleon of a nucleus varies with